LEARNING HUB MASTER SERIES • PHYSICS VOLUME 3

Physics: Work, Energy & Momentum

Scalar Products & Line Integrals • The Work-Kinetic Energy Theorem • Conservative vs. Non-Conservative Forces • Potential Energy Landscapes & Equilibrium Stability • Conservation of Mechanical Energy • Linear Momentum & Impulse • Center of Mass & Variable-Mass Rocket Propulsion • 1D / 2D Collisions & Fast-Track Hand Calculation Shortcuts

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Table of Contents

Part I: Work, Line Integrals & The Work-Kinetic Energy Theorem

Chapter 1: Definition of Work, Scalar Dot Product & Variable Force Line Integrals ⚡ FAST-TRACK W1 & W2

Chapter 2: The Work-Kinetic Energy Theorem (1D & 3D Derivations) & Power

Part II: Conservative Forces & Potential Energy Landscapes

Chapter 3: Conservative vs. Non-Conservative Forces, Path Independence & The Curl Test ⚡ FAST-TRACK W1

Chapter 4: Potential Energy Curves U(x), Turning Points & Equilibrium Stability ⚡ FAST-TRACK W3

Chapter 5: Conservation of Mechanical Energy, Dissipative Work & Fluid Drag Dynamics

Part III: Linear Momentum, Impulse & System Dynamics

Chapter 6: Linear Momentum, Newton's Second Law & The Impulse-Momentum Theorem

Chapter 7: Center of Mass Motion, König's Theorem & Rocket Propulsion (Variable Mass)

Part IV: Collisions in One and Two Dimensions

Chapter 8: 1D Collisions, Coefficient of Restitution (e) & Center-of-Mass Frame Shortcuts ⚡ FAST-TRACK W4 & W5

Chapter 9: 2D Oblique Collisions, CM Scattering Geometry & 90° Orthogonality ⚡ FAST-TRACK W6

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Rigorous Worked Problems (Parameters, Derivations, Units & Sanity Checks)

Appendix: Master Work, Energy & Momentum Formula Reference Sheet

Part I: Work, Line Integrals & The Work-Kinetic Energy Theorem

Chapter 1: Definition of Work & Variable Force Line Integrals

In classical Newtonian mechanics, force describes the instantaneous interaction between objects. However, tracking motion over extended displacements is dramatically simplified by introducing scalar energy quantities. **Work** quantifies the cumulative action of a force acting over a spatial path.

1.1 Constant Force & The Vector Dot Product

When a constant force **F** acts on a point particle undergoing a rectilinear displacement **s**, the work done is defined as the scalar (dot) product:

$$W = \mathbf{F} \cdot \mathbf{s} = F\, s \cos\theta = F_{\parallel}\, s$$

where θ is the angle between the force vector **F** and the displacement vector **s**, and $F_{\parallel} = F \cos\theta$ represents the component of force acting parallel to the trajectory. In SI units, work is measured in **Joules (J)**, where 1 J = 1 N·m = 1 $kg \cdot m^2/s^2$.

> **Geometric Properties of the Work Dot Product**
>
> • **Positive Work ($0° \le \theta < 90°$):** Force has a component in the direction of motion, pumping kinetic energy into the particle ($\cos\theta > 0$).
> • **Zero Work ($\theta = 90°$):** Normal forces, centripetal forces, and magnetic Lorentz forces act perpendicularly to displacement ($\cos 90° = 0$); they alter direction but do no work.
> • **Negative Work ($90° < \theta \le 180°$):** Force opposes motion (e.g., kinetic friction, viscous drag), removing mechanical energy from the particle ($\cos 180° = -1$).

1.2 Variable Force in 1D and the Riemann Definite Integral

When the force component F_x varies continuously as a function of position x, we partition the displacement from x_i to x_f into infinitesimal segments dx. Over each infinitesimal displacement, the force is quasi-constant, doing infinitesimal work $dW = F_x(x)\, dx$. Summing these contributions yields the definite Riemann line integral:

$$W = \int_{x_i}^{x_f} F_x(x)\, dx$$

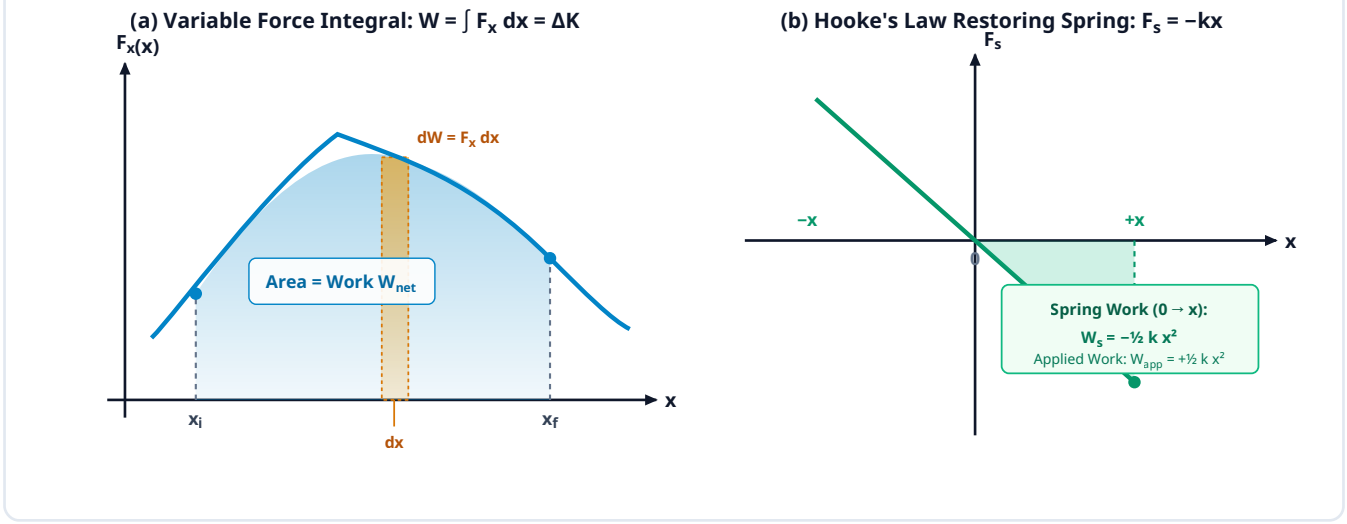


Figure 3.1: (a) Work as the Riemann definite area under a variable force curve $F_x(x)$. (b) Work done by a linear Hooke's law spring restoring force $F_s = -kx$.

1.3 Work Done by a Linear Spring (Hooke's Law)

An ideal spring exerts an elastic restoring force described by Hooke's law: $F_s(x) = -k\,x$, where x is displacement from the equilibrium configuration (x = 0) and k is the spring stiffness constant (N/m). The work done by the spring as the block moves from x_i to x_f is:

$$W_s = \int_{x_i}^{x_f} (-k\,x)\,dx = -k\,\frac{1}{2}\,[x^2]_{x_i}^{x_f} = \frac{1}{2}\,k\,x_i^2 - \frac{1}{2}\,k\,x_f^2$$

Notice that when stretching the spring from rest ($x_i = 0$ to $x_f = x$), the spring does negative work $W_s = -\tfrac{1}{2}kx^2$ because the restoring force opposes displacement. An external applied agent pulling the spring quasi-statically exerts $F_{app} = +kx$ and performs positive work $W_{app} = +\tfrac{1}{2}kx^2$.

⚡ Fast-Track W2: Spring Work via Average Linear Force

Because Hooke's force varies linearly with displacement from 0 to x, the average force is simply the arithmetic mean of initial and final values:

$$F_{avg} = \frac{F(0) + F(x)}{2} = \frac{0 + k\,x}{2} = \frac{1}{2}\,k\,x \quad \Rightarrow \quad W = F_{avg} \cdot x = \frac{1}{2}\,k\,x^2$$

This 5-second mental calculation bypasses integration entirely for any linear force profile!

1.4 General 3D Curvilinear Line Integral

For a particle traveling along an arbitrary three-dimensional trajectory Γ parameterized from point $\mathbf{r}_1$ to $\mathbf{r}_2$ under a spatially varying force field $\mathbf{F}(\mathbf{r})$, the total work is given by the contour line integral:

$$W = \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_{x_1}^{x_2} F_x\,dx + \int_{y_1}^{y_2} F_y\,dy + \int_{z_1}^{z_2} F_z\,dz$$

Chapter 2: The Work-Kinetic Energy Theorem & Power

The Work-Kinetic Energy Theorem is the direct bridge between Newton's second law of motion and energy conservation. It states that the net work performed by all forces acting on a particle equals the change in its translational kinetic energy.

The Work-Kinetic Energy Theorem

$$W_{net} = \Delta K = K_f - K_i = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

This fundamental theorem holds universally for both constant and variable forces, across straight and curved trajectories, and in both inertial and non-inertial frames (when fictitious inertial forces are properly included).

Rigorous Calculus Derivation (1D & 3D)

Starting from Newton's second law $F_{net} = m\,a = m\,\frac{dv}{dt}$, and applying the chain rule of calculus $\frac{dv}{dt} = \frac{dv}{dx}\,\frac{dx}{dt} = v\,\frac{dv}{dx}$:

$$W_{net} = \int_{x_i}^{x_f} F_{net}\,dx = \int_{x_i}^{x_f} m\,v\,\frac{dv}{dx}\,dx = \int_{v_i}^{v_f} m\,v\,dv = m\left[\frac{v^2}{2}\right]_{v_i}^{v_f} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

In three dimensions, using vector dot products: $\mathbf{F}_{net} \cdot d\mathbf{r} = m\,\frac{d\mathbf{v}}{dt} \cdot \mathbf{v}\,dt = m\,\mathbf{v} \cdot d\mathbf{v} = \frac{1}{2} m\,d(\mathbf{v} \cdot \mathbf{v}) = \frac{1}{2} m\,d(v^2)$.

Integrating from state 1 to state 2 yields exactly ΔK. □

2.1 Instantaneous and Average Power

Power defines the rate of energy transfer or work delivery over time.

$$P_{avg} = \frac{\Delta W}{\Delta t}, \qquad P(t) = \frac{dW}{dt} = \frac{\mathbf{F} \cdot d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v}$$

In SI units, power is measured in **Watts (W)** (1 W = 1 J/s = 1 kg·m^2/s^3). In engineering, mechanical power is often expressed in horsepower (hp), where 1 hp = 745.7 W = 550 ft·lb/s.

Part II: Conservative Forces & Potential Energy Landscapes

Chapter 3: Conservative vs. Non-Conservative Forces & The Curl Test

Forces occurring in nature divide into two fundamental classes based on path dependence:

Property	Conservative Forces	Non-Conservative Forces
Path Dependence	Work is strictly independent of the trajectory taken between endpoints.	Work depends explicitly on the path length and geometry.
Closed Loop Integral	$\oint \mathbf{F} \cdot d\mathbf{r} = 0$ (Zero net work around any closed contour).	$\oint \mathbf{F} \cdot d\mathbf{r} \neq 0$ (Energy is irreversibly dissipated or supplied).
Potential Function	Exists: $\mathbf{F} = -\nabla U$ $\quad (\Delta U = -W_c)$.	No scalar potential energy function can be defined.
Primary Examples	Gravity ($\mathbf{F} = -mg\,\hat{\mathbf{j}}$), Hooke's spring ($F = -kx$), Electrostatic (Coulomb).	Kinetic friction ($f_k = \mu_k N$), fluid drag, propulsion thrust.

⚡ Fast-Track W1: The 3D Curl Test for Conservative Forces

To mathematically test if an analytical force field $\mathbf{F}(x, y, z) = F_x\,\hat{\mathbf{i}} + F_y\,\hat{\mathbf{j}} + F_z\,\hat{\mathbf{k}}$ is conservative on a simply connected domain, compute its curl:

$$\nabla \times \mathbf{F} = \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} = 0, \quad \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = 0, \quad \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 0$$

If and only if $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, the force is conservative! Then work is simply $W = U(\mathbf{r}_i) - U(\mathbf{r}_f)$, requiring no path integration.

Chapter 4: Potential Energy Landscapes U(x) & Equilibrium Stability

For a conservative force in one dimension, the potential energy function U(x) is defined such that:

$$F_x(x) = -\frac{dU}{dx} \quad \Leftrightarrow \quad \Delta U = U(x_f) - U(x_i) = -\int_{x_i}^{x_f} F_x(x)\,dx$$

The minus sign indicates that physical systems naturally accelerate toward regions of **lower potential energy**.

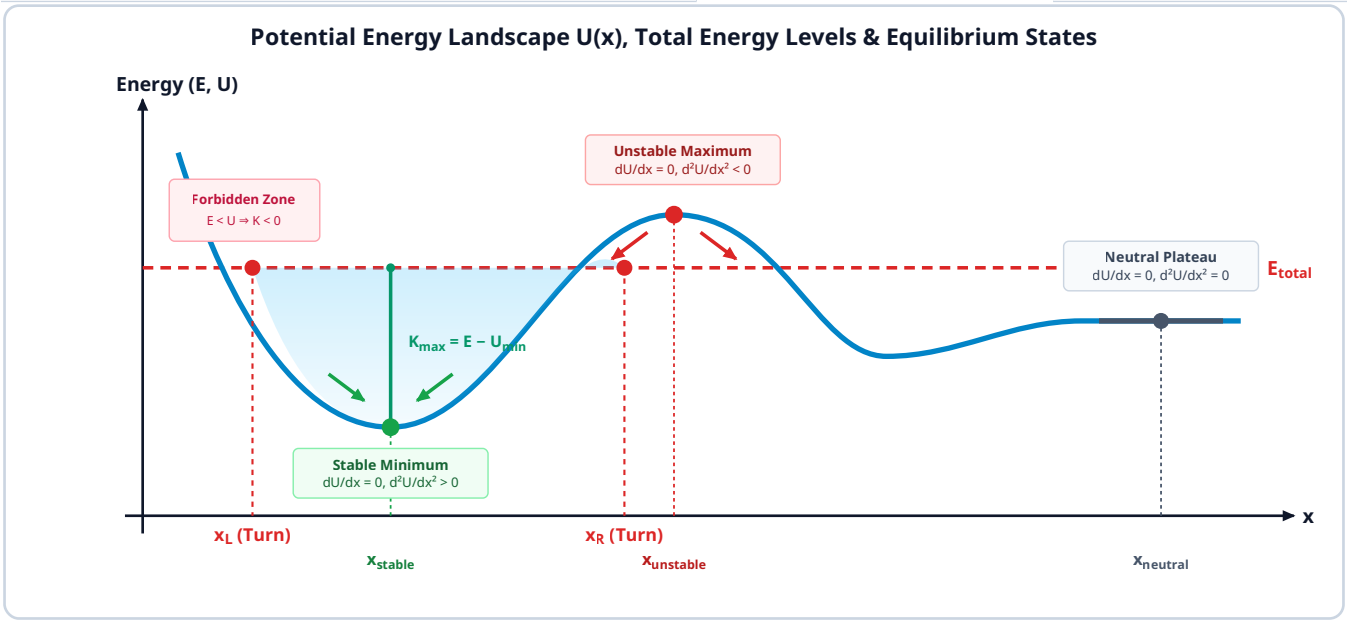


Figure 3.2: General Potential Energy Landscape U(x) showing total mechanical energy E_{total}, turning points x_L, x_R, kinetic energy head K(x), and equilibrium classifications.

4.1 Turning Points and Classically Allowed Regions

For an isolated particle moving under conservative forces, the total mechanical energy is constant: E = K(x) + U(x). Because kinetic energy must be non-negative ($K = \frac{1}{2}mv^2 \geq 0$), the particle's motion is strictly constrained to the **classically allowed region**:

$$U(x) \leq E \quad \Rightarrow \quad K(x) = E - U(x) \geq 0, \qquad v(x) = \pm \sqrt{[\, 2(E - U(x)) / m \,]}$$

Points where U(x) = E are called **turning points** (x_L and x_R). At these points, K = 0, speed momentarily vanishes, and the force $F_x = -dU/dx$ accelerates the particle back into the interior of the well.

4.2 Equilibrium Classifications via Taylor Derivatives

Equilibrium occurs where the net force vanishes: $F_x = -dU/dx = 0$ (the slope of U(x) is horizontal). The second derivative d^2U/dx^2 determines the stability:

Equilibrium Type	Calculus Condition	Force Profile under Displacement (Δx)	Physical Behavior
Stable Equilibrium	dU/dx = 0, **$d^2U/dx^2 > 0$** (Local Minimum)	Restoring force: $F \approx -k_{eff}\,\Delta x$	Particle oscillates stably about the bottom of the well.
Unstable Equilibrium	dU/dx = 0, **$d^2U/dx^2 < 0$** (Local Maximum)	Repulsive force: $F \approx +k_{eff}\,\Delta x$	Any displacement triggers runaway divergence away from x_0.
Neutral Equilibrium	dU/dx = 0, **$d^2U/dx^2 = 0$** (Flat Plateau)	Zero force: F = 0 for small Δx	Particle remains at rest in its new displaced position.

⚡ Fast-Track W3: Small Oscillations Frequency via Potential Curvature

Expanding U(x) in a Taylor series about a stable minimum x_0:

$$U(x) \approx U(x_0) + \frac{1}{2} U''(x_0)(x - x_0)^2 \quad \Rightarrow \quad k_{eff} = \frac{d^2U}{dx^2}_{x=x_0}$$

The natural angular frequency of small vibrations is instantly obtained by inspection:

$$\omega = \sqrt{\frac{k_{eff}}{m}} = \sqrt{\frac{U''(x_0)}{m}}$$

Chapter 5: Conservation of Mechanical Energy & Dissipative Work

The total work performed on a system is the sum of conservative work W_c and non-conservative work W_{nc}:

$$W_{net} = W_c + W_{nc} = \Delta K$$

Substituting the definition of potential energy $W_c = -\Delta U$ into the theorem yields the master formulation of energy conservation:

$$W_{nc} = \Delta K + \Delta U = \Delta E_{mech} = (K_f + U_f) - (K_i + U_i)$$

Conservation of Mechanical Energy

If all active forces are conservative ($W_{nc} = 0$), mechanical energy is strictly conserved:

$$E_{mech} = K + U_{grav} + U_{spring} = \text{constant} \quad \Rightarrow \quad K_i + U_i = K_f + U_f$$

When friction or drag is present, $W_{nc} = -f_k\, d = -\Delta E_{th}$, converting ordered mechanical energy into random microscopic thermal energy ($\Delta E_{total} = \Delta E_{mech} + \Delta E_{th} = 0$).

5.1 Dissipative Work: Friction vs. Viscous and Aerodynamic Drag

Non-conservative forces remove macroscopic mechanical energy from a moving body and convert it irreversibly into internal thermal energy. The three standard physical dissipation regimes are:

• **Coulomb Sliding Friction:** Constant magnitude $f_k = \mu_k N$ opposing instantaneous velocity. The work done over total path length s is always negative: $W_{fric} = -\mu_k N\, s$.

• **Linear Viscous Drag (Low Reynolds Number):** For small particles in laminar fluids, $\mathbf{F}_d = -b\,\mathbf{v}$. The terminal fall velocity under gravity is $v_t = m\,g / b$.

• **Quadratic Aerodynamic Drag (High Reynolds Number):** For macroscopic bodies in air, $F_d = \tfrac{1}{2} C_D \rho A v^2$. The terminal velocity occurs when drag balances gravity: $v_t = \sqrt{[\, 2\, m\, g / (\rho\, A\, C_D)\,]}$. At terminal velocity, the rate of mechanical energy loss equals the power delivered by gravity: $P_{diss} = F_d\, v_t = m\, g\, v_t$.

Part III: Linear Momentum, Impulse & System Dynamics

Chapter 6: Linear Momentum & The Impulse-Momentum Theorem

For a particle of mass m moving with velocity **v**, its **linear momentum** is the vector quantity:

$$\mathbf{p} = m\,\mathbf{v} \qquad [\text{kg·m/s} = \text{N·s}]$$

Newton originally formulated his second law in terms of momentum flux: the net force equals the time rate of change of linear momentum:

$$\mathbf{F}_{net} = \frac{d\mathbf{p}}{dt}$$

6.1 The Impulse-Momentum Theorem

Integrating Newton's second law over a finite collision time interval from t_i to t_f defines **impulse J**:

$$\mathbf{J} = \int_{t_i}^{t_f} \mathbf{F}(t)\,dt = \mathbf{F}_{avg}\,\Delta t = \Delta\mathbf{p} = \mathbf{p}_f - \mathbf{p}_i$$

During short-duration impacts (e.g., bat hitting a baseball, vehicle collision), the impact force is enormous compared to external forces like gravity or friction. Under the **impulse approximation**, external non-impact forces are neglected during the collision interval Δt.

Chapter 7: Center of Mass Motion, System Dynamics & Rocket Propulsion

For an N-particle discrete system with total mass $M = \sum m_i$, the **Center of Mass (CM)** position vector is:

$$\mathbf{R}_{cm} = \frac{1}{M}\sum_{i=1}^{N} m_i\,\mathbf{r}_i = \frac{1}{M}\int \mathbf{r}\,dm$$

Differentiating with respect to time yields the velocity of the center of mass:

$$\mathbf{V}_{cm} = \frac{d\mathbf{R}_{cm}}{dt} = \frac{1}{M}\sum_{i=1}^{N} m_i\,\mathbf{v}_i = \frac{\mathbf{P}_{total}}{M}$$

Newton's Second Law for a Multi-Particle System

Summing forces across all particles: by Newton's third law, internal action-reaction pairs cancel identically ($\sum \mathbf{F}_{int} = \mathbf{0}$). Consequently:

$$\mathbf{F}_{net,\,ext} = M\,\mathbf{A}_{cm} = \frac{d\mathbf{P}_{total}}{dt}$$

Conservation of Linear Momentum: If the net external force on a system is zero ($\mathbf{F}_{net,\,ext} = \mathbf{0}$), the total linear momentum is strictly conserved: $\mathbf{P}_{total}$ = constant, and $\mathbf{V}_{cm}$ = constant.

7.1 Variable Mass Systems: The Tsiolkovsky Rocket Equation

A rocket accelerates by ejecting propellant backward at constant exhaust speed v_{ex} relative to the rocket body. Applying momentum conservation to the system (rocket + ejected propellant) between time t and t + dt:

$$m\frac{dv}{dt} = -v_{ex}\frac{dm}{dt} \quad \Rightarrow \quad dv = -v_{ex}\frac{dm}{m}$$

Integrating from initial launch mass m_0 to burnout mass m_f yields the celebrated **Tsiolkovsky Rocket Equation**:

$$\Delta v = v_f - v_i = v_{ex}\ln\frac{m_0}{m_f}$$

The rocket thrust is given by $F_{thrust} = |dm/dt|\,v_{ex}$.

7.2 König's Theorem: Kinetic Energy Decomposition in the CM Frame

A fundamental theorem in analytical mechanics relates the total kinetic energy of any multi-particle system to its center-of-mass motion and internal motion relative to the CM frame:

$$K_{total} = K_{cm} + K_{rel} = \frac{1}{2}M\,V_{cm}^{2} + \sum_{i=1}^{N}\frac{1}{2}m_i\,\mathbf{v}_i^{*2}$$

where $\mathbf{v}_i^* = \mathbf{v}_i - \mathbf{V}_{cm}$ is the velocity of particle i in the CM frame. The cross term $\sum m_i\,\mathbf{V}_{cm}\cdot\mathbf{v}_i^*$ vanishes identically because $\sum m_i\,\mathbf{v}_i^* = \mathbf{0}$ by definition of the center of mass!

Physical Consequence of König's Theorem in Collisions

In any isolated collision without external forces, $\mathbf{V}_{cm}$ is strictly constant; therefore, $K_{cm} = \tfrac{1}{2}M\,V_{cm}^{2}$ is totally invariant and completely unavailable for dissipation! Only the internal relative kinetic energy K_{rel} can be converted into heat, acoustic waves, or permanent plastic deformation. In a completely inelastic collision (e = 0), exactly all of K_{rel} is annihilated, leaving $K_{final} = K_{cm}$.

Part IV: Collisions in One and Two Dimensions

Chapter 8: 1D Collisions & The Center-of-Mass Frame Shortcut

In any isolated collision, linear momentum is conserved: $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$. Collisions are classified by the degree of kinetic energy preservation through the **coefficient of restitution** e:

$$e = \frac{\text{Relative Speed of Recession}}{\text{Relative Speed of Approach}} = \frac{v_{2f} - v_{1f}}{v_{1i} - v_{2i}}$$

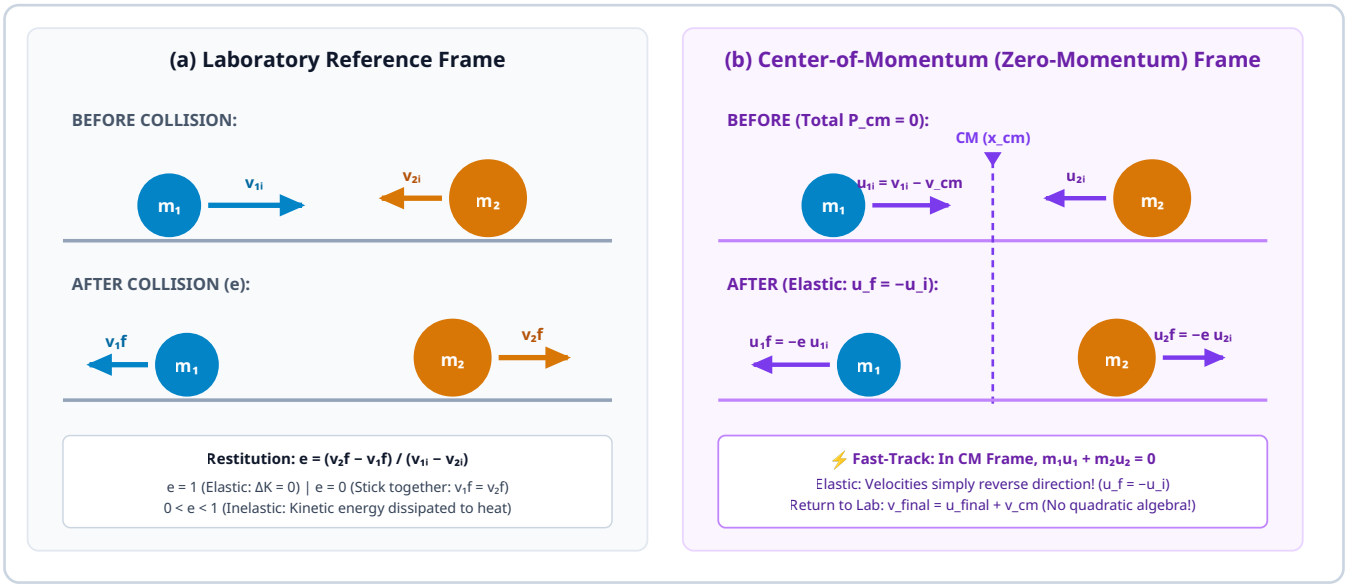


Figure 3.3: 1D Collision Kinematics: (a) Laboratory reference frame. (b) Center-of-Momentum (Zero-Momentum) frame where elastic collisions reduce to simple velocity reflections.

8.1 Collision Classification Spectrum

Type	Restitution (e)	Kinetic Energy (ΔK)	Post-Collision Dynamics
Perfect Elastic	e = 1.0	ΔK = 0 (Conserved)	Relative speed of recession equals speed of approach.
Inelastic	0 < e < 1.0	ΔK < 0 (Dissipated)	Energy partially converted to heat, sound, or permanent deformation.
Completely Inelastic	e = 0.0	Maximum Possible Loss	Objects stick together, moving at identical final velocity $v_f = v_{cm}$.
Superelastic (Explosive)	e > 1.0	ΔK > 0 (Gained)	Internal chemical or spring potential energy released into kinetic motion.

⚡ Fast-Track W5: The Center-of-Mass Frame Reflection Shortcut

Standard textbook derivations of 1D elastic collisions require tedious simultaneous quadratic equations. The **Center-of-Mass (CM) method** eliminates quadratic algebra completely in 3 simple steps:

1. **Compute CM Velocity:** $v_{cm} = (m_1 v_{1i} + m_2 v_{2i}) / (m_1 + m_2)$.
2. **Transform to CM Frame:** $u_{1i} = v_{1i} - v_{cm}$, $u_{2i} = v_{2i} - v_{cm}$.
3. **Reflect Velocities:** In CM frame, elastic collision simply negates velocities: $u_{1f} = -u_{1i}$, $u_{2f} = -u_{2i}$ (or multiply by $-e$ if inelastic!).
4. **Return to Lab Frame:** $v_{1f} = u_{1f} + v_{cm}$, $v_{2f} = u_{2f} + v_{cm}$.

This shortcut delivers the exact post-collision speeds in under 30 seconds with zero risk of algebraic error!

⚡ Fast-Track W4: Kinetic Energy Loss in Completely Inelastic Collisions

When a projectile m_1 collides completely inelastically ($e = 0$) with a stationary target m_2 ($v_{2i} = 0$), the fraction of initial kinetic energy lost to deformation and heat is:

$$\frac{\Delta K_{loss}}{K_{initial}} = \frac{m_2}{m_1 + m_2} \quad \text{and} \quad \frac{K_{final}}{K_{initial}} = \frac{m_1}{m_1 + m_2}$$

Rule of Thumb: If a light bullet strikes a heavy block ($m_2 \gg m_1$), virtually 100% of kinetic energy is lost. If a heavy truck strikes a stationary pebble ($m_1 \gg m_2$), virtually 0% of kinetic energy is lost!

Chapter 9: 2D Oblique Collisions & Momentum Vector Geometry

In two-dimensional collisions, momentum conservation is applied independently along orthogonal Cartesian axes:

$$\sum p_{xi} = \sum p_{xf} \quad \Rightarrow \quad m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$
$$\sum p_{yi} = \sum p_{yf} \quad \Rightarrow \quad m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

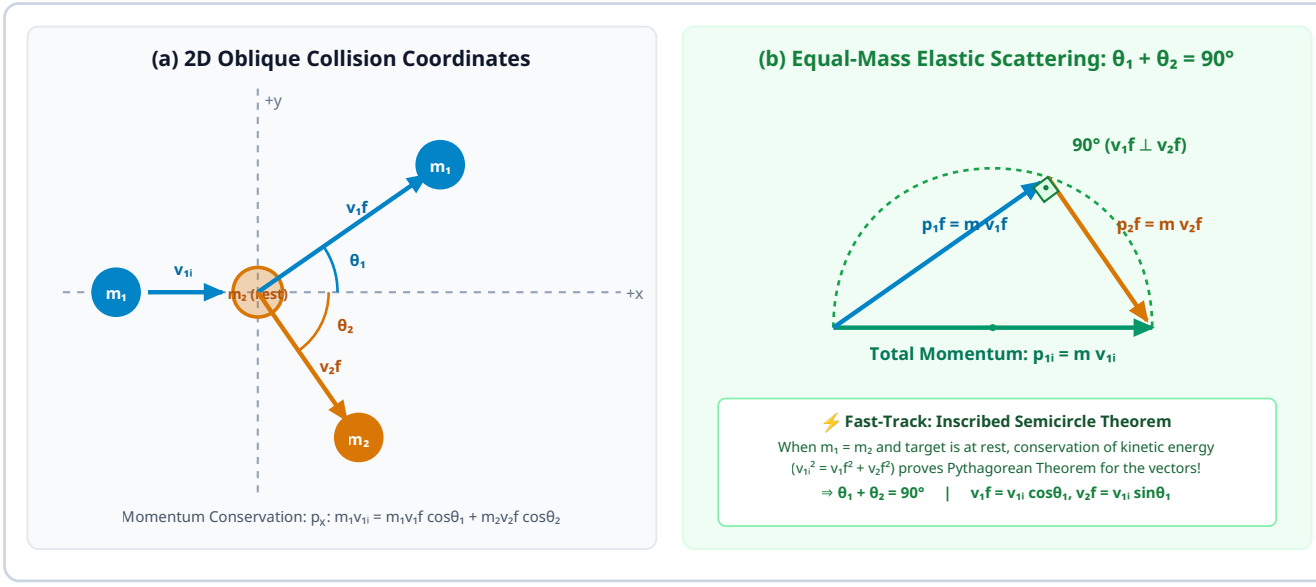


Figure 3.4: (a) 2D Oblique Collision coordinate geometry. (b) Momentum vector addition triangle and Thales' 90° orthogonality circle for identical masses in elastic scattering.

⚡ Fast-Track W6: Equal-Mass Elastic Scattering 90° Orthogonality

When a particle of mass m strikes an identical stationary particle of mass m elastically:

$$\mathbf{v}_{1i} = \mathbf{v}_{1f} + \mathbf{v}_{2f} \quad \text{(Momentum)} \quad \text{and} \quad v_{1i}^2 = v_{1f}^2 + v_{2f}^2 \quad \text{(Kinetic Energy)}$$

Squaring the momentum vector: $\mathbf{v}_{1i} \cdot \mathbf{v}_{1i} = v_{1f}^2 + v_{2f}^2 + 2\,\mathbf{v}_{1f} \cdot \mathbf{v}_{2f}$. Equating with kinetic energy proves:

$$2\,\mathbf{v}_{1f} \cdot \mathbf{v}_{2f} = 0 \quad \Rightarrow \quad \mathbf{v}_{1f} \perp \mathbf{v}_{2f} \quad \Rightarrow \quad \theta_1 + \theta_2 = 90^\circ$$

The post-collision trajectories form a right triangle inscribed in a semicircle! Post-collision speeds are immediately obtained by simple trigonometry: $\mathbf{v_{1f} = v_{1i} \cos \theta_1}$ and $\mathbf{v_{2f} = v_{1i} \sin \theta_1}$.

9.1 Center-of-Momentum Frame Analysis in 2D Collisions

In the Center-of-Momentum (CM) frame, the total momentum is identically zero both before and after collision: $\mathbf{p}_1^* = -\mathbf{p}_2^*$. Consequently, in an elastic collision, the particles approach collinearly and recede collinearly at an angle θ^* with their momentum magnitudes completely unchanged:

$$|\mathbf{p}_{1f}^*| = |\mathbf{p}_{1i}^*|, \qquad |\mathbf{p}_{2f}^*| = |\mathbf{p}_{2i}^*|$$

The scattering process in the CM frame is nothing more than a rigid rotation of the momentum axis by the scattering angle θ^*! The relationship between the laboratory scattering angle θ_1 and the CM scattering angle θ^* is given by:

$$\tan \theta_1 = \frac{\sin \theta^*}{\cos \theta^* + (m_1 / m_2)}$$

When $m_1 = m_2$, this yields $\tan \theta_1 = \sin \theta^* / (1 + \cos \theta^*) = \tan(\theta^* / 2)$, proving $\theta_1 = \theta^* / 2$.

9.2 Impact Parameter and Hard-Sphere Geometric Scattering

For two smooth, spherical particles of radii R_1 and R_2 colliding with an impact parameter b (the perpendicular distance between their initial velocity lines):

$$b = (R_1 + R_2) \sin \alpha \quad \Rightarrow \quad \theta^* = \pi - 2\,\alpha = \pi - 2 \arcsin \frac{b}{R_1 + R_2}$$

If $b = 0$, the collision is perfectly head-on ($\alpha = 0$, $\theta^* = 180^\circ$). If $b = R_1 + R_2$, the spheres merely graze each other ($\alpha = 90^\circ$, $\theta^* = 0^\circ$).

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Complete Worked Problems with Fast-Tracks

Problem 1: Variable Force Line Integral and Work-Kinetic Energy Theorem

Chapters 1, 2 • Work-KE Theorem

Problem Statement: A block of mass m = 2.0 kg is pushed across a rough horizontal floor by a position-dependent horizontal force $F(x) = (3x^2 - 4x + 6)$ N, where x is in meters. The block moves from $x_i = 1.0$ m to $x_f = 4.0$ m. The coefficient of kinetic friction between the block and floor is $\mu_k = 0.20$, and the initial speed at x_i is $v_i = 3.0$ m/s. Calculate: (1) work done by the applied force W_F, (2) work done by friction W_{fric}, and (3) final speed of the block v_f at $x_f = 4.0$ m.

1. Given Parameters & Knowns:

m = 2.0 kg, $x_i = 1.0$ m, $x_f = 4.0$ m, $v_i = 3.0$ m/s, $\mu_k = 0.20$, $g = 9.80\ m/s^2$.

2. Symbolic Algebraic Derivation:

The applied force does work via the Riemann integral: $W_F = \int_{x_i}^{x_f} F(x)\,dx$.

Normal force is N = m g, giving a constant kinetic friction force $f_k = \mu_k m g$ opposing displacement: $W_{fric} = -f_k \Delta x = -\mu_k m g (x_f - x_i)$.

By the Work-Kinetic Energy Theorem: $W_{net} = W_F + W_{fric} = \Delta K = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$.

Solving for final speed: $v_f = \sqrt{[\, v_i^2 + 2 W_{net} / m \,]}$.

3. Numerical Calculation with Explicit SI Units:

• Applied Work: $W_F = \int_{1.0}^{4.0} (3x^2 - 4x + 6)\,dx = [x^3 - 2x^2 + 6x]_{1.0}^{4.0}$.

At x = 4.0: $(4)^3 - 2(4)^2 + 6(4) = 64 - 32 + 24 = 56.0$ J.

At x = 1.0: $(1)^3 - 2(1)^2 + 6(1) = 1 - 2 + 6 = 5.0$ J.

$\Rightarrow W_F = 56.0 - 5.0 =$ **51.0 J**.

• Friction Work: $W_{fric} = -(0.20)(2.0\ kg)(9.80\ m/s^2)(4.0\ m - 1.0\ m) = -(3.92\ N)(3.0\ m) =$ **−11.76 J**.

• Net Work: $W_{net} = 51.0\ J + (-11.76\ J) =$ **39.24 J**.

• Initial Kinetic Energy: $K_i = \frac{1}{2}(2.0\ kg)(3.0\ m/s)^2 = 9.0$ J.

• Final Kinetic Energy: $K_f = 9.0\ J + 39.24\ J = 48.24$ J.

• Final Velocity: $v_f = \sqrt{[\, 2(48.24\ J) / (2.0\ kg) \,]} = \sqrt{48.24} \approx$ **6.95 m/s**.

4. Physical Sanity & Boundary Checks:

✓ **Energy Inflow:** $W_{net} > 0$, which correctly accelerates the block from 3.0 m/s to 6.95 m/s.

✓ **Frictionless Limit:** If $\mu_k \to 0$, $W_{net} = 51.0$ J $\Rightarrow v_f = \sqrt{[9.0 + 51.0]} = \sqrt{60} \approx 7.75$ m/s > 6.95 m/s, consistent with friction dissipating 11.76 J.

Problem 2: Lennard-Jones Potential Landscape, Equilibrium & Small Vibrations

Chapter 4 • Fast-Track W3

Problem Statement: The interaction potential between two atoms in a diatomic molecule is modeled by the Lennard-Jones potential $U(r) = U_0 \left[(r_0/r)^{12} - 2(r_0/r)^6 \right]$, where r is the internuclear separation, $U_0 = 4.0 \times 10^{-21}$ J, and $r_0 = 0.38$ nm. (1) Find the equilibrium bond length r_{eq} and well depth. (2) Compute the effective spring constant k_{eff} and small-vibration frequency ω for reduced mass $\mu = 1.66 \times 10^{-27}$ kg. (3) Determine turning points for total energy $E = -0.50\ U_0$.

1. Given Parameters & Knowns:

$U_0 = 4.0 \times 10^{-21}$ J, $r_0 = 0.38 \times 10^{-9}$ m, $\mu = 1.66 \times 10^{-27}$ kg, $E = -2.0 \times 10^{-21}$ J.

2. Symbolic Algebraic Derivation:

Equilibrium: $dU/dr = 0 \Rightarrow U_0 \left[-12\, r_0^{12}/r^{13} + 12\, r_0^6/r^7 \right] = 0 \Rightarrow (r_0/r)^6 = 1 \Rightarrow r_{eq} = r_0$.

Well depth: $U(r_0) = U_0 \left[1 - 2 \right] = -U_0$.

Curvature & stiffness: $k_{eff} = d^2U/dr^2 \big|_{r_0} = U_0 \left[156\, r_0^{12}/r^{14} - 84\, r_0^6/r^8 \right] \big|_{r_0} = 72\, U_0 / r_0^2$.

Turning points: $U(r) = E \Rightarrow (r_0/r)^{12} - 2(r_0/r)^6 = E / U_0$. Setting $y = (r_0/r)^6$: $y^2 - 2y - (E/U_0) = 0$.

3. Numerical Calculation with Explicit SI Units:

• Equilibrium Distance: $r_{eq} = \mathbf{0.38\ nm}$; Minimum Energy: $U_{min} = \mathbf{-4.0 \times 10^{-21}\ J}$.

• Effective Stiffness: $k_{eff} = 72\ (4.0 \times 10^{-21}\ \text{J}) / (0.38 \times 10^{-9}\ \text{m})^2 = 2.88 \times 10^{-19} / 1.444 \times 10^{-19} = \mathbf{1.99\ N/m}$.

• Vibration Frequency (via W3): $\omega = \sqrt{(1.9945 / 1.66 \times 10^{-27})} = \sqrt{(1.201 \times 10^{27})} \approx \mathbf{1.10 \times 10^{12}\ rad/s}$ ($f \approx 175$ GHz).

• Turning Points: $y^2 - 2y + 0.50 = 0 \Rightarrow y = 1 \pm \sqrt{(1 - 0.50)} = 1 \pm 0.7071$.

$y_1 = 1.7071 \Rightarrow r_L = r_0 / (1.7071)^{1/6} = 0.38\ \text{nm} / 1.0932 \approx \mathbf{0.348\ nm}$.

$y_2 = 0.2929 \Rightarrow r_R = r_0 / (0.2929)^{1/6} = 0.38\ \text{nm} / 0.8146 \approx \mathbf{0.466\ nm}$.

4. Physical Sanity & Boundary Checks:

✓ **Turning Point Straddling:** r_L (0.348 nm) < r_{eq} (0.380 nm) < r_R (0.466 nm). Motion is strictly bounded.

✓ **Kinetic Head:** At equilibrium, $K_{max} = E - U(r_0) = -0.50\ U_0 - (-U_0) = +0.50\ U_0 > 0$, which is physically consistent.

Problem 3: The Ballistic Pendulum & Completely Inelastic Energy Dissipation

Chapters 5, 6, 8 • Fast-Track W4

Problem Statement: A ballistic pendulum consists of a wooden block of mass M = 3.50 kg suspended by a light rigid rod of length L = 1.80 m. A high-speed bullet of mass m = 12.0 g is fired horizontally into the block and embeds itself inside it. Following the collision, the pendulum swings upward through a maximum angle θ = 35.0°. Calculate: (1) the speed V of the combined system immediately after impact, (2) the initial muzzle velocity v_0 of the bullet, and (3) the exact percentage of mechanical kinetic energy lost in the collision.

1. Given Parameters & Knowns:

m = 0.0120 kg, M = 3.50 kg, L = 1.80 m, θ = 35.0°, g = 9.80 m/s^2.

2. Symbolic Algebraic Derivation:

• **Phase 1 (Collision):** Extremely rapid collision; gravity impulse is negligible. Momentum is strictly conserved: m v_0 = (m + M) V ⇒ v_0 =

$$\frac{m + M}{m} V.$$

• **Phase 2 (Swing):** Cord tension does no work. Mechanical energy is conserved during the swing: ½ (m + M) V^2 = (m + M) g h, where h = L(1 − cos θ). Therefore: V = √[2 g L (1 − cos θ)].

• **Energy Loss:** By Fast-Track W4, fractional kinetic energy lost is ΔK / K_i = M / (m + M).

3. Numerical Calculation with Explicit SI Units:

• Height Rise: h = 1.80 m × (1 − cos 35.0°) = 1.80 m × (1 − 0.81915) = 1.80 × 0.18085 = **0.3255 m**.

• Post-Impact Speed: V = √[2(9.80 m/s^2)(0.3255 m)] = √6.3804 = **2.526 m/s**.

• Initial Bullet Speed: v_0 =

$$\frac{0.0120\ \text{kg} + 3.50\ \text{kg}}{0.0120\ \text{kg}} \times 2.526\ \text{m/s} = \frac{3.512}{0.0120} \times 2.526 = 292.67 \times 2.526 = \textbf{739 m/s}.$$

• Energy Accounting: K_i = ½ (0.0120 kg)(739.3 m/s)^2 = **3,279 J**.

K_f = ½ (3.512 kg)(2.526 m/s)^2 = **11.2 J**.

• Energy Dissipation: ΔK_{loss} = 3,279 − 11.2 = 3,267.8 J ⇒ **99.66% dissipated into heat**.

Fast-Track W4 Check: M / (m + M) = 3.50 / 3.512 = **99.66%**. Identical result!

4. Physical Sanity & Boundary Checks:

✓ **Muzzle Velocity Order of Magnitude:** 739 m/s corresponds to standard military/rifle ballistics (700–800 m/s).

✓ **Mass Ratio Invariance:** The block is ~292 times heavier than the bullet; thus ~99.7% of initial energy must be converted to internal deformation/heat during the inelastic embedding process.

Problem 4: 1D Inelastic Collision with Restitution (e) & CM Transformation

Chapter 8 • Fast-Track W5

Problem Statement: On a frictionless horizontal air track, Glider 1 of mass m_1 = 0.50 kg moves with initial velocity v_{1i} = +4.00 m/s toward Glider 2 of mass m_2 = 1.00 kg moving with v_{2i} = −2.00 m/s. Calculate the final velocities v_{1f} and v_{2f} using the Center-of-Mass Frame Shortcut for: (1) perfectly elastic collision (e = 1.0), and (2) partially inelastic collision with coefficient of restitution e = 0.60. Verify momentum and kinetic energy conservation.

1. Given Parameters & Knowns:

m_1 = 0.50 kg, m_2 = 1.00 kg, v_{1i} = +4.00 m/s, v_{2i} = −2.00 m/s.

2. Symbolic Algebraic Derivation (Fast-Track W5):

Step 1: Compute Center of Mass Velocity: $v_{cm} = (m_1 v_{1i} + m_2 v_{2i}) / (m_1 + m_2)$.

Step 2: Transform velocities to CM frame: $u_{1i} = v_{1i} - v_{cm}$, $u_{2i} = v_{2i} - v_{cm}$.

Step 3: Apply Restitution in CM: $u_{1f} = -e\, u_{1i}$, $u_{2f} = -e\, u_{2i}$.

Step 4: Revert to Laboratory Frame: $v_{1f} = u_{1f} + v_{cm}$, $v_{2f} = u_{2f} + v_{cm}$.

3. Numerical Calculation with Explicit SI Units:

• Center of Mass Speed: v_{cm} = [(0.50 kg)(4.00 m/s) + (1.00 kg)(−2.00 m/s)] / (0.50 + 1.00) = (2.00 − 2.00) / 1.50 = **0.00 m/s**!

Because v_{cm} = 0, the lab frame IS the Zero-Momentum CM frame!

• Initial CM velocities: u_{1i} = +4.00 m/s, u_{2i} = −2.00 m/s.

• **Case 1: Elastic (e = 1.0):**

$v_{1f} = -u_{1i}$ = **−4.00 m/s**, $v_{2f} = -u_{2i}$ = **+2.00 m/s**.

Initial KE: K_i = ½(0.50)(4.00)^2 + ½(1.00)(−2.00)^2 = 4.00 + 2.00 = **6.00 J**.

Final KE: K_f = ½(0.50)(−4.00)^2 + ½(1.00)(+2.00)^2 = 4.00 + 2.00 = **6.00 J**. (ΔK = 0 ✓).

• **Case 2: Inelastic (e = 0.60):**

v_{1f} = −(0.60)(+4.00 m/s) = **−2.40 m/s**.

v_{2f} = −(0.60)(−2.00 m/s) = **+1.20 m/s**.

Momentum Check: p_f = (0.50)(−2.40) + (1.00)(1.20) = −1.20 + 1.20 = **0.00 kg·m/s** (✓ Conserved).

Final KE: K_f = ½(0.50)(−2.40)^2 + ½(1.00)(1.20)^2 = 1.44 + 0.72 = **2.16 J**.

Energy loss: ΔK = 6.00 − 2.16 = 3.84 J (64% lost, exactly matching 1 − e^2 = 1 − 0.36 = 0.64 ✓).

4. Physical Sanity & Boundary Checks:

✓ **Speed of Separation:** For e = 0.60: $v_{2f} - v_{1f}$ = 1.20 − (−2.40) = 3.60 m/s. Relative approach was 4.00 − (−2.00) = 6.00 m/s. Ratio is 3.60 / 6.00 = 0.60 = e.

✓ **No Overlap:** Glider 1 moves left at 2.40 m/s, Glider 2 moves right at 1.20 m/s; they recede smoothly with no spatial interference.

Problem 5: 2D Oblique Billiard Scattering & 90° Orthogonality Invariant

Chapter 9 • Fast-Track W6

Problem Statement: Cue ball A of mass m = 0.170 kg moving at speed v_0 = 5.00 m/s along the +x axis strikes an identical target ball B (m = 0.170 kg) initially at rest on a frictionless billiard table. Following the elastic collision, ball A deflects at an angle θ_A = 30.0° above the original line of motion. Calculate: (1) the scattering angle θ_B of ball B, (2) the final speeds v_{Af} and v_{Bf}, and (3) verify momentum and kinetic energy conservation.

1. Given Parameters & Knowns:

$m_A = m_B = m = 0.170$ kg, $v_{Ai} = 5.00$ m/s, $v_{Bi} = 0$, $\theta_A = 30.0°$, elastic (e = 1.0).

2. Symbolic Algebraic Derivation:

By Fast-Track W6, an elastic collision between identical masses with target at rest guarantees orthogonal scattering trajectories:

$$\theta_A + \theta_B = 90.0° \quad \Rightarrow \quad \theta_B = 90.0° - \theta_A$$

The vector triangle formed by $\mathbf{v}_0 = \mathbf{v}_{Af} + \mathbf{v}_{Bf}$ is a right triangle with hypotenuse v_0:

$v_{Af} = v_0 \cos\theta_A$, $v_{Bf} = v_0 \sin\theta_A$.

3. Numerical Calculation with Explicit SI Units:

• Scattering Angle of B: $\theta_B = 90.0° - 30.0° =$ **60.0° below the +x axis** (−60.0°).

• Final Speed of Ball A: $v_{Af} = (5.00\text{ m/s}) \cos 30.0° = 5.00 \times 0.86603 =$ **4.33 m/s**.

• Final Speed of Ball B: $v_{Bf} = (5.00\text{ m/s}) \sin 30.0° = 5.00 \times 0.50000 =$ **2.50 m/s**.

• **Momentum Conservation Verification:**

x-component: $p_{xf} = (0.170)[\,4.330 \cos 30° + 2.500 \cos(-60°)\,] = 0.170\,[\,3.750 + 1.250\,] = 0.170 \times 5.00 =$ **0.850 kg·m/s** $= p_{xi}$ (✓).

y-component: $p_{yf} = (0.170)[\,4.330 \sin 30° + 2.500 \sin(-60°)\,] = 0.170\,[\,2.165 - 2.165\,] =$ **0.000 kg·m/s** $= p_{yi}$ (✓).

• **Kinetic Energy Verification:**

$K_i = \tfrac{1}{2}(0.170\text{ kg})(5.00\text{ m/s})^2 =$ **2.125 J**.

$K_f = \tfrac{1}{2}(0.170)[\,(4.330)^2 + (2.500)^2\,] = \tfrac{1}{2}(0.170)[\,18.75 + 6.25\,] = \tfrac{1}{2}(0.170)(25.0) =$ **2.125 J** (✓).

4. Physical Sanity & Boundary Checks:

✓ **Head-on Limit ($\theta_A \to 0°$):** $v_{Af} \to 0$, $v_{Bf} \to v_0$; incident ball stops completely and transfers 100% momentum to target.

✓ **Glancing Limit ($\theta_A \to 90°$):** $v_{Af} \to v_0$, $v_{Bf} \to 0$; incident ball skims past target unaffected.

Appendix: Master Work, Energy & Momentum Formula Reference Sheet

Concept / Domain	Master Equation	Operational Context & SI Units
Work (Constant Force)	$W = \mathbf{F} \cdot \mathbf{s} = F s \cos\theta$	Joules [J = N·m = kg·m^2/s^2]
Work (Variable Force Line Integral)	$W = \int_\Gamma \mathbf{F} \cdot d\mathbf{r} = \int (F_x\,dx + F_y\,dy + F_z\,dz)$	General 3D curvilinear path Γ
Hooke's Law Spring Work	$W_s = \frac{1}{2} k x_i^2 - \frac{1}{2} k x_f^2$	Restoring force: $F_s = -kx$
Work-Kinetic Energy Theorem	$W_{net} = \Delta K = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$	Universal for all net forces
Instantaneous Power	$P = dW/dt = \mathbf{F} \cdot \mathbf{v}$	Watts [W = J/s]; 1 hp = 745.7 W
Potential Energy Gradient	$\mathbf{F} = -\nabla U \quad \Leftrightarrow \quad F_x = -\partial U/\partial x$	Conservative force field condition
Curl Test for Conservatism	$\nabla \times \mathbf{F} = \mathbf{0} \quad \Leftrightarrow \quad \oint \mathbf{F} \cdot d\mathbf{r} = 0$	Independent of path in simply connected space
Mechanical Energy Conservation	$\Delta E_{mech} = \Delta K + \Delta U = W_{nc}$	$W_{nc} = -f_k\,d$ (Dissipation by friction)
Small Oscillations Frequency	$\omega = \sqrt{[\,U''(x_0) / m\,]}$	Taylor curvature at stable minimum x_0
Impulse-Momentum Theorem	$\mathbf{J} = \int \mathbf{F}(t)\,dt = \mathbf{F}_{avg}\,\Delta t = \Delta\mathbf{p}$	Linear momentum: $\mathbf{p} = m\,\mathbf{v}$ [N·s]
Center of Mass Motion	$\mathbf{R}_{cm} = \frac{\sum m_i \mathbf{r}_i}{M}, \quad \mathbf{F}_{ext} = M\,\mathbf{A}_{cm} = \frac{d\mathbf{P}_{tot}}{dt}$	Internal forces cancel by Newton's 3rd Law
König's Kinetic Energy Theorem	$K_{tot} = \frac{1}{2} M V_{cm}^2 + \sum \frac{1}{2} m_i v_i^{*2} = K_{cm} + K_{rel}$	Cross-term vanishes identically
Tsiolkovsky Rocket Equation	$\Delta v = v_{ex} \ln(m_0 / m_f)$	Thrust: $F_{th} = \lvert dm/dt \rvert\, v_{ex}$
Coefficient of Restitution	$e = (v_{2f} - v_{1f}) / (v_{1i} - v_{2i})$	e = 1 (Elastic), e = 0 (Completely inelastic)
CM Frame Collision Reflection	$u_{1f} = -e\,u_{1i}, \quad u_{2f} = -e\,u_{2i}$	Lab return: $v_f = u_f + v_{cm}$
2D Equal-Mass Elastic Scattering	$\theta_1 + \theta_2 = 90°, \quad v_{1f} = v_{1i} \cos\theta_1$	Target at rest; Thales' semicircle invariant